WORKED SOLUTION

STEP II 1996, Question 8

Pure • Second-Order Differential Equations

> **STEP II 1996, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/96-s2-q8

The key idea

The equation mixes $\mathrm{f}(x)$ with $\mathrm{f}(-x)$, so it isn't an ordinary differential equation for f at all. The trick is to write the equation down twice, once at x and once at $-x$, and then add and subtract. The sum g and the difference h each satisfy an honest constant-coefficient equation that we can solve with a complementary function and a particular integral, and f is recovered as $\frac{1}{2}(\mathrm{g}+\mathrm{h})$.

Solution

The values of $\mathrm{g}(0)$ and $\mathrm{g}'(0)$

Putting $x=0$ in $\mathrm{g}(x)=\mathrm{f}(x)+\mathrm{f}(-x)$ gives $\mathrm{g}(0)=2\mathrm{f}(0)=2$.

Differentiating, with the chain rule giving a minus sign on the second term,

$$\mathrm{g}'(x)=\mathrm{f}'(x)-\mathrm{f}'(-x),$$

so $\mathrm{g}'(0)=\mathrm{f}'(0)-\mathrm{f}'(0)=0$, as required. (Notice that we didn't need the value of $\mathrm{f}'(0)$: g is an even function, so its derivative is odd and vanishes at 0.)

The equation for g

Differentiating again, the two minus signs from the chain rule cancel:

$$\mathrm{g}''(x)=\mathrm{f}''(x)+\mathrm{f}''(-x).$$

? **Ask yourself** What is $\mathrm{f}''(-x)$? The given equation only tells us about $\mathrm{f}''(x)$.

The given equation holds for every x, so we may replace x by $-x$ throughout. Since cos is even, this gives

$$\mathrm{f}''(-x)+\mathrm{f}(x)=-x+3\cos 2x.$$

Adding this to the original equation,

$$\mathrm{f}''(x)+\mathrm{f}''(-x)+\mathrm{f}(-x)+\mathrm{f}(x)=6\cos 2x, \qquad \text{that is,} \qquad \mathrm{g}''(x)+\mathrm{g}(x)=6\cos 2x,$$

as required.

Solving for g

The auxiliary equation $m^2+1=0$ has roots $\pm\mathrm{i}$, so the complementary function is $A\cos x+B\sin x$. For a particular integral we try $\mathrm{g}=k\cos 2x$: then $\mathrm{g}''+\mathrm{g}=-4k\cos 2x+k\cos 2x=-3k\cos 2x$, so $k=-2$. Hence

$$\mathrm{g}(x)=A\cos x+B\sin x-2\cos 2x.$$

Now $\mathrm{g}(0)=A-2=2$ gives $A=4$, and $\mathrm{g}'(0)=B=0$. So

$$\mathrm{g}(x)=4\cos x-2\cos 2x.$$

Solving for h

For $\mathrm{h}(x)=\mathrm{f}(x)-\mathrm{f}(-x)$ we have

$$\mathrm{h}(0)=0, \qquad \mathrm{h}'(x)=\mathrm{f}'(x)+\mathrm{f}'(-x), \quad \text{so} \quad \mathrm{h}'(0)=2\mathrm{f}'(0)=-2,$$

and $\mathrm{h}''(x)=\mathrm{f}''(x)-\mathrm{f}''(-x)$. This time we subtract the equation at $-x$ from the equation at x:

$$\mathrm{f}''(x)-\mathrm{f}''(-x)+\mathrm{f}(-x)-\mathrm{f}(x)=2x, \qquad \text{that is,} \qquad \mathrm{h}''(x)-\mathrm{h}(x)=2x.$$

? **Ask yourself** Why has the sign in front of h changed from + to −?

In the subtraction the f terms appear as $\mathrm{f}(-x)-\mathrm{f}(x)$, which is $-\mathrm{h}(x)$, not $+\mathrm{h}(x)$. So the solutions are now exponential rather than trigonometric: the auxiliary equation $m^2-1=0$ gives the complementary function $C\mathrm{e}^{x}+D\mathrm{e}^{-x}$. Trying $\mathrm{h}=px+q$ as a particular integral gives $-(px+q)=2x$, so $p=-2$, $q=0$. Hence

$$\mathrm{h}(x)=C\mathrm{e}^{x}+D\mathrm{e}^{-x}-2x.$$

The conditions give $\mathrm{h}(0)=C+D=0$ and $\mathrm{h}'(0)=C-D-2=-2$, so $C=D=0$ and

$$\mathrm{h}(x)=-2x.$$

Recovering f

Adding the definitions of g and h gives $\mathrm{g}+\mathrm{h}=2\mathrm{f}$, so

$$\mathrm{f}(x)=\tfrac{1}{2}(\mathrm{g}(x)+\mathrm{h}(x))=\tfrac{1}{2}(4\cos x-2\cos 2x-2x)=2\cos x-\cos 2x-x,$$

which is the required result.

Checks

- $\mathrm{f}(0) = 2 - 1 - 0 = 1$ and $\mathrm{f}'(x) = -2\sin x + 2\sin 2x - 1$, so $\mathrm{f}'(0) = -1$.
- Substituting back: $\mathrm{f}''(x) = -2\cos x + 4\cos 2x$ and $\mathrm{f}(-x) = 2\cos x - \cos 2x + x$, which add to $x + 3\cos 2x$.
- The parities are right: g came out even and h odd, as they must.
- Stepping the pair $\mathrm{f}(x)$, $\mathrm{f}(-x)$ numerically from $x = 0$ to $x = 3$ gives $\mathrm{f}(3) \approx -5.940155$, agreeing with $2\cos 3 - \cos 6 - 3$.

Common slips

- Forgetting the chain-rule sign: $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{f}(-x) = -\mathrm{f}'(-x)$, but $\frac{\mathrm{d}^2}{\mathrm{d}x^2}\mathrm{f}(-x) = +\mathrm{f}''(-x)$.
- Replacing x by $-x$ on only one side of the equation: the x on the right becomes $-x$ too.
- Getting $\mathrm{h}'' + \mathrm{h} = 2x$ and ending up with trigonometric terms in h.

Beyond STEP

This is a functional differential equation, in which the unknown is evaluated at more than one point. Splitting into even and odd parts works because reflection $x \mapsto -x$ is a linear map that squares to the identity, so its eigenvalues are ± 1 and every function splits uniquely into an even part (eigenvalue 1) and an odd part (eigenvalue -1). On each part the equation becomes an ordinary one. This idea of decomposing by symmetry runs all through linear algebra and Fourier analysis. For the constant-coefficient equations themselves, see Boyce and DiPrima, *Elementary Differential Equations*; for the linear-algebra viewpoint, Strang, *Introduction to Linear Algebra*.

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